

BASIC ELECTRONICS

(24B11EC111)

SOLUTION TUTORIAL-8

Answer.1

(a)

$$\begin{aligned}V_{T0} &= \Phi_{GC} - 2\phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}} \\ \phi_F &= \frac{kT}{q} \ln \frac{n_i}{N_A} = 0.026 \ln \frac{1.45 \times 10^{10}}{2 \times 10^{15}} = -0.308 \text{ V} \\ C_{ox} &= \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.97 \times 8.854 \times 10^{-14}}{200 \times 10^{-8}} = 1.758 \times 10^{-7} \text{ F/cm}^2 \\ Q_{B0} &= -\sqrt{2qN_A\epsilon_{si}|-2\phi_F|} \\ &= -\sqrt{2 \cdot 1.6 \times 10^{-19} \cdot 2 \times 10^{15} \cdot 11.7 \times 8.854 \times 10^{-14} \cdot |-2 \times 0.308|} \\ &= -2.01 \times 10^{-8} \\ V_{T0} &= -0.85 + 0.615 + \frac{2 \times 10^{-8}}{1.758 \times 10^{-7}} - \frac{1.6 \times 10^{-19} \cdot 2 \times 10^{11}}{1.758 \times 10^{-7}} = -0.303 \text{ V}\end{aligned}$$

(b) p-type implanted needed in the amount of:

$$\begin{aligned}\Delta V &= 0.8 - V_{T0} = 0.8 + 0.303 = 1.303 = \frac{qN_I}{C_{ox}} \\ N_I &= \frac{1.303 C_{ox}}{q} = \frac{1.303 \cdot 1.758 \times 10^{-7}}{1.6 \times 10^{-19}} = 1.21 \times 10^{12} \text{ cm}^{-2}\end{aligned}$$

Answer.2

$$\begin{aligned}\phi_F(\text{substrate}) &= \frac{kT}{q} \ln \frac{N_{D,sub}}{n_i} = 0.026 \ln \frac{1 \times 10^{16}}{1.45 \times 10^{10}} = 0.348 \text{ [V]} \\ \phi_F(\text{gate}) &= \frac{kT}{q} \ln \frac{N_{D,poly}}{n_i} = 0.026 \ln \frac{1 \times 10^{20}}{1.45 \times 10^{10}} = 0.587 \text{ [V]} \\ \Phi_{GC} &= \phi_F(\text{substrate}) - \phi_F(\text{gate}) = 0.348 - 0.587 = -0.239 \text{ [V]} \\ C_{ox} &= \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.9 \times 8.85 \times 10^{-14}}{0.1 \times 10^{-4}} = 3.45 \times 10^{-8} \text{ [F/cm}^2\text{]}\end{aligned}$$

$$\begin{aligned}
Q_{B0} &= \sqrt{2qN_{D,sub}\epsilon_{si}|2\phi_F|} \\
&= \sqrt{2 \times 1.6 \times 10^{-19} \times 10^{16} \times 11.7 \times 8.85 \times 10^{-14} \times 2 \times 0.348} \\
&= 4.8 \times 10^{-8} [C/cm^2]
\end{aligned}$$

$$\begin{aligned}
V_{T0} &= \Phi_{GC} - 2\phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}} \\
&= -0.239 - 2 \times 0.348 - \frac{4.8 \times 10^{-8}}{3.45 \times 10^{-8}} - \frac{4 \times 10^{10} \times 1.6 \times 10^{-19}}{3.45 \times 10^{-8}} \\
&= -2.51 [V]
\end{aligned}$$

Answer.3.

(a) For unimplanted transistor,

$$\phi_F(\text{substrate}) = \frac{kT}{q} \ln \frac{n_i}{N_A} = 0.026 \ln \frac{1.45 \times 10^{10}}{1.0 \times 10^{16}} = -0.35 [V]$$

$$\phi_F(\text{gate}) = \frac{kT}{q} \ln \frac{N_{D,poly}}{n_i} = 0.026 \ln \frac{1 \times 10^{20}}{1.45 \times 10^{10}} = 0.59 [V]$$

$$\Phi_{GC} = \phi_F(\text{substrate}) - \phi_F(\text{gate}) = -0.35 - 0.59 = -0.94 [V]$$

$$\begin{aligned}
Q_{B0} &= \sqrt{2qN_{A,sub}\epsilon_{si}|2\phi_F|} \\
&= \sqrt{2 \times 1.6 \times 10^{-19} \times 10^{16} \times 11.7 \times 8.85 \times 10^{-14} \times 2 \times 0.35} \\
&= -4.82 \times 10^{-8} [C/cm^2]
\end{aligned}$$

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.9 \times 8.85 \times 10^{-14}}{500 \times 10^{-8}} = 6.9 \times 10^{-8} [F/cm^2]$$

$$\begin{aligned}
V_{T0} &= \Phi_{GC} - 2\phi_F - \frac{Q_{B0}}{C_{ox}} - \frac{Q_{ox}}{C_{ox}} \\
&= -0.94 - 2 \times (-0.35) - \frac{(-4.82 \times 10^{-8})}{6.9 \times 10^{-8}} - \frac{2 \times 10^{10} \times 1.6 \times 10^{-19}}{6.9 \times 10^{-8}} \\
&= 0.41 [V]
\end{aligned}$$

(b) For $V_T = 2V$:

$$V_T = 2 = V_{T0} - \frac{Q_{II}}{C_{ox}} = 0.412 - \frac{Q_{II}}{C_{ox}}$$

Negative charges needed in this case, so it must be p-type implant in the amount of

$$Q_{II} = qN_I = (V_T - V_{T0})C_{ox}$$

$$N_I = (2 - 0.41) \times \frac{6.9 \times 10^{-8}}{1.6 \times 10^{-19}} = 6.85 \times 10^{11} [cm^{-3}]$$

For $V_T = -2V$, positive charges need, must be n-type implant,

$$N_I = (2 + 0.41) \times \frac{6.9 \times 10^{-8}}{1.6 \times 10^{-19}} = 1.04 \times 10^{12} [cm^{-3}]$$

Answer. 4

(a) Find V_{T0}

For both case 1 (Row1) and case 2(Row2), the transistor operates in saturation region, using equation (3.62)

$$I_D(sat) = \frac{k_n}{2} (V_{GS} - V_{T0})^2$$

$$\frac{I_D(Row1)}{I_D(Row2)} = \frac{(4 - V_{T0})^2}{(5 - V_{T0})^2} = \frac{256}{441}$$

$$\frac{4 - V_{T0}}{5 - V_{T0}} = \pm \frac{16}{21}$$

The correct solution is:

$$V_{T0} = 0.8[V]$$

(b) Find mobility

$$C_{ox} = \frac{\epsilon_{ox}}{t_{ox}} = \frac{3.9 \times 8.85 \times 10^{-14}}{345 \times 10^{-8}} = 1.0 \times 10^{-7} [F/cm]$$

$$256 \times 10^{-6} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (4 - V_{T0})^2$$

$$\mu_n = \frac{2 \times 256 \times 10^{-6}}{1.0 \times 10^{-7} \times 1.0 \times 3.2^2} = 500 [cm^2/V \cdot s]$$

(c) Find body effect coefficient γ

Using either Row 3 or Row4 data and equation (3.62), first find V_T

$$144 \times 10^{-6} = \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (4 - V_T)^2$$

$$V_T = 1.6[V]$$

$$\gamma = \frac{V_T(V_{SB}) - V_{T0}}{\sqrt{|2\phi_F| + V_{SB}} - \sqrt{|2\phi_F|}} = \frac{1.6 - 0.8}{\sqrt{0.64 + 2.6} - \sqrt{0.64}} = 0.8 [V^{1/2}]$$

Answer 5

As a rule of thumb for designing this classical biasing circuit, we choose R_D and R_S to provide one-third of the power-supply voltage V_{DD} as a drop across each of R_D , the transistor (i.e., V_{DS}), and R_S . For $V_{DD} = 15$ V, this choice makes $V_D = +10$ V and $V_S = +5$ V. Now, since I_D is required to be 0.5 mA, we can find the values of R_D and R_S as follows:

$$R_D = \frac{V_{DD} - V_D}{I_D} = \frac{15 - 10}{0.5} = 10 \text{ k}\Omega$$

$$R_S = \frac{V_S}{I_D} = \frac{5}{0.5} = 10 \text{ k}\Omega$$

The required value of V_{GS} can be determined by first calculating the overdrive voltage V_{OV} from

$$I_D = \frac{1}{2} k'_n (W/L) V_{OV}^2$$

$$0.5 = \frac{1}{2} \times 1 \times V_{OV}^2$$

which yields $V_{OV} = 1$ V, and thus,

$$V_{GS} = V_t + V_{OV} = 1 + 1 = 2 \text{ V}$$

Now, since $V_S = +5$ V, V_G must be

$$V_G = V_S + V_{GS} = 5 + 2 = 7 \text{ V}$$

To establish this voltage at the gate we may select $R_{G1} = 8 \text{ M}\Omega$ and $R_{G2} = 7 \text{ M}\Omega$. The final circuit is shown in Fig. 7.49. Observe that the dc voltage at the drain (+10 V) allows for a positive signal swing of +5 V (i.e., up to V_{DD}) and a negative signal swing of 4 V [i.e., down to $(V_G - V_t)$].

If the NMOS transistor is replaced with another having $V_t = 1.5$ V, the new value of I_D can be found as follows:

$$I_D = \frac{1}{2} \times 1 \times (V_{GS} - 1.5)^2$$

$$V_G = V_{GS} + I_D R_S$$

$$7 = V_{GS} + 10 I_D$$

$$I_D = 0.455 \text{ mA}$$

Thus the change in I_D is

$$\Delta I_D = 0.455 - 0.5 = -0.045 \text{ mA}$$

which is $\frac{-0.045}{0.5} \times 100 = -9\%$ change.